Exp No:10 Experiment to understand array function in Data science.

AIM:

To understand array function in Data science.

Description:

Understand array function using Numpy library

CODE:

```
import numpy as np
array = np.array([15, 15, 16, 16, 17, 17, 18, 18, 19])
print("1. Original Array:")
print(array)
print("\n2. Square Root of Array:")
print(np.sqrt(array))
print("\n3. Number of Dimensions:")
print(array.ndim)
new_array = array.reshape(3, 3)
print("\n4. Reshaped Array:")
print(new_array)
print("\n5. Number of Dimensions after Reshape:")
print(new_array.ndim)
print("\n6. Ravelled Array:")
print(new_array.ravel())
newm = new_array.reshape(3, 3)
print("\n7. New Array:")
print(newm)
print("\n8. newm[2,1:3]:")
print(newm[2,1:3])
print("\n9. newm[1:2,1:3]:")
print(newm[1:2,1:3])
```

```
print("\n10. new_array[0:3,0:0]:")
print(new_array[0:3,0:0])
print("\n11. new_array[0:2,0:1]:")
print(new_array[0:2,0:1])
print("\n12. new_array[0:3,0:1]:")
print(new_array[0:3,0:1])
print("\n13. new_array[1:3]:")
print(new_array[1:3])
```

OUTPUT:

```
1. Original Array:
[15 15 16 16 17 17 18 18 19]
2. Square Root of Array:
[3.87298335 3.87298335 4.        4.        4.12310563 4.12310563
 4.24264069 4.24264069 4.35889894]
3. Number of Dimensions:
1
4. Reshaped Array:
[[15 15 16]
 [16 17 17]
 [18 18 19]]
5. Number of Dimensions after Reshape:
2
6. Ravelled Array:
[15 15 16 16 17 17 18 18 19]
7. New Array:
[[15 15 16]
 [16 17 17]
```

[18 18 19]]

8. newm[2,1:3]:

[18 19]

9. newm[1:2,1:3]:

[[17 17]]

10. new_array[0:3,0:0]:

[]

11. new_array[0:2,0:1]:

[[15]

[16]]

12. new_array[0:3,0:1]:

[[15]

[16]

[18]]

13. new_array[1:3]:

[[16 17 17]

[18 18 19]]

RESULT:

The array operations ere successfully performed using numpy including reshaping,indexing,slicing,ravelling.